

■ Times Tables 2 & 5 ■

Magic Patterns — Full Lesson Guide

Grade / Year	Grade 2 (ages 7–8)
Subject	Mathematics — Multiplication
Session	Session 1 of 4
Duration	45–60 minutes
Academy	PLANRS ACADEMY
Prepared by	Class Teacher

Learning Objectives

- Recall multiplication facts for the 2 and 5 times tables up to 10.
- Recognise and explain the even-number pattern in the 2× table.
- Recognise the 0-or-5 ending pattern in the 5× table.
- Apply times-table knowledge to real-life word problems.
- Understand that multiplication is commutative (e.g. $2 \times 5 = 5 \times 2$).

Materials Needed

- Whiteboard / projector with slides
- Printed number cards (2–20 even; 5, 10, 15 ... 50)
- Pairs of objects for warm-up (socks, shoes, gloves, bangles)
- Counters / coins for grouping activities
- Student workbooks & pencils
- Optional: clapping sticks or desk drums for chanting rhythm

1. Warm-Up: The Pair Game (5 min)

Activate prior knowledge of pairs and even numbers before introducing multiplication.

Step	Teacher Action	Student Action
1	Hold up pairs of objects one at a time (shoes, socks, gloves...)	Count the total aloud together
2	Ask: "How many pairs? How many objects in total?"	Call out answers; show on fingers
3	Write on board: 3 pairs → $3 \times 2 = ?$	Predict the answer before calculating
4	Reveal: $3 \times 2 = 6$. Discuss what pattern they notice.	Notice: totals are always even

■ *Teacher tip: Use real objects from the classroom — shoes are always a crowd-pleaser!*

2. Building the 2 Times Table (10 min)

Concept: Repeated Addition → Multiplication

$2 + 2 = 4 \rightarrow 2 \times 2 = 4$

$2 + 2 + 2 = 6 \rightarrow 3 \times 2 = 6$

$2 + 2 + 2 + 2 = 8 \rightarrow 4 \times 2 = 8$

Emphasise: **multiplying by 2 is the same as doubling**. Have students use their fingers: one hand for each group of 2.

Full 2 Times Table Reference

$2 \times 1 = 2$	$2 \times 2 = 4$	$2 \times 3 = 6$	$2 \times 4 = 8$	$2 \times 5 = 10$
$2 \times 6 = 12$	$2 \times 7 = 14$	$2 \times 8 = 16$	$2 \times 9 = 18$	$2 \times 10 = 20$

Magic Patterns in the 2× Table

- ✓ All answers are **EVEN** numbers.
- ✓ Answers always end in 0, 2, 4, 6, or 8.
- ✓ Each answer is **2 more** than the previous one.
- ✓ Multiplying by 2 = **doubling** the number.

Chant (clap the rhythm together):

*"Two, four, six, eight — who do we appreciate? Ten, twelve, fourteen — let's keep the pattern clean!
Sixteen, eighteen, twenty — two-times tables — we've got plenty!"*

2× Table in Everyday Life

Example	Working	Answer
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Riya has 3 pairs of bangles	3×2	6 bangles
4 auto-rickshaws, each has 2 mirrors	4×2	8 mirrors
Amma buys 5 biscuit packets, 2 rows each	5×2	10 rows
6 cricket innings, 2 openers each	6×2	12 openers

3. Building the 5 Times Table (10 min)

Concept: Hands & Fingers

Start by having students hold up their hands and count in fives:

1 hand = 5 fingers → $1 \times 5 = 5$

2 hands = 10 fingers → $2 \times 5 = 10$

3 hands = 15 fingers → $3 \times 5 = 15$

...continue up to 10 hands = 50 fingers → $10 \times 5 = 50$

Full 5 Times Table Reference

$5 \times 1 = 5$	$5 \times 2 = 10$	$5 \times 3 = 15$	$5 \times 4 = 20$	$5 \times 5 = 25$
$5 \times 6 = 30$	$5 \times 7 = 35$	$5 \times 8 = 40$	$5 \times 9 = 45$	$5 \times 10 = 50$

Magic Patterns in the 5× Table

- ✓ Every answer ends in **0 or 5**.
- ✓ Numbers increase by **5** each time.
- ✓ Odd multiples ($\times 1$, $\times 3$, $\times 5$...) end in **5**.
- ✓ Even multiples ($\times 2$, $\times 4$, $\times 6$...) end in **0**.
- ✓ The 5× table matches the **minute hand** on a clock!

Chant:

"Five, ten, fifteen, twenty — count by fives, we've got plenty! Twenty-five, thirty, thirty-five — counting fives keeps maths alive! Forty, forty-five, and fifty — five-times tables, aren't they nifty!"

■ *Teacher tip: Say $5 \times 5 = 25$ and $5 \times 10 = 50$ extra loud — students love it!*

5× Table in Everyday Life

Example	Working	Answer
5 students each have a ■5 coin	5×5	■25
Laddoo box: 3 rows of 5 laddoos	3×5	15 laddoos
6 classrooms, each has 5 windows	6×5	30 windows
4 rows of Diwali diyas, 5 per row	4×5	20 diyas

4. Commutativity Discovery (5 min)

Write on the board and ask students what they notice:

$2 \times 5 = 10$

$5 \times 2 = 10$

$2 \times 7 = 14$

$7 \times 2 = 14$

Key teaching point: The *order* of numbers in multiplication does not matter — the answer is always the same. This is called the **commutative property**.

■ Ask: "If you know $2 \times 8 = 16$, do you also know 8×2 ?" Celebrate when students realise they get two facts for the price of one!

5. Quick-Fire Drill (5 min)

Call out the following in rapid succession. Students respond by calling out the answer together. Keep the energy high!

Question	Answer	Question	Answer
$2 \times 8 = ?$	16	$5 \times 3 = ?$	15
$2 \times 5 = ?$	10	$5 \times 5 = ?$	25
$2 \times 9 = ?$	18	$5 \times 7 = ?$	35
$2 \times 6 = ?$	12	$5 \times 9 = ?$	45
$2 \times 10 = ?$	20	$5 \times 6 = ?$	30

6. Spot the Error! (5 min)

Read out each "wrong" answer. Students shout "NO!" and give the correct answer.

Student says...	Why it's wrong	Correct answer
$2 \times 9 = 19$	The $2\times$ table always gives even numbers; 19 is odd	$2 \times 9 = 18$
$5 \times 6 = 32$	$5\times$ table answers must end in 0 or 5; 32 ends in 2	$5 \times 6 = 30$
$2 \times 7 = 15$	15 is odd — impossible for $2\times$ table	$2 \times 7 = 14$
$5 \times 4 = 22$	22 doesn't end in 0 or 5	$5 \times 4 = 20$

7. Guided Practice — Fill in the Blanks (8 min)

2× Table Practice

$2 \times 3 = \underline{\quad}$	$2 \times 7 = \underline{\quad}$	$2 \times \underline{\quad} = 18$	$\underline{\quad} \times 2 = 12$
$2 \times 4 = \underline{\quad}$	$2 \times 9 = \underline{\quad}$	$2 \times \underline{\quad} = 14$	$\underline{\quad} \times 2 = 20$

5× Table Practice

$5 \times 4 = \underline{\quad}$	$5 \times 8 = \underline{\quad}$	$5 \times \underline{\quad} = 35$	$\underline{\quad} \times 5 = 45$
$5 \times 3 = \underline{\quad}$	$5 \times 6 = \underline{\quad}$	$5 \times \underline{\quad} = 25$	$\underline{\quad} \times 5 = 50$

Answers — 2×: 6, 14, 9, 6 | 8, 18, 7, 10

Answers — 5×: 20, 40, 7, 9 | 15, 30, 5, 10

8. Word Problems (7 min)

Solve together as a class, encouraging students to write the multiplication sentence first.

1 .	Priya has 4 pairs of socks. How many socks in total?	$4 \times 2 = 8$ socks
2 .	6 cricket teams each have 5 fielders on the boundary. How many fielders?	$6 \times 5 = 30$ fielders
3 .	Amma arranges 7 biscuit packets, each with 2 rows. How many rows?	$7 \times 2 = 14$ rows
4 .	8 students each save $\pounds 5$ a day. How much do they save together?	$8 \times 5 = \pounds 40$
5 .	There are 9 auto-rickshaws. Each has 2 mirrors. How many mirrors in all?	$9 \times 2 = 18$ mirrors
6 .	A diya stand has 6 rows with 5 diyas each. How many diyas?	$6 \times 5 = 30$ diyas

9. Recap & Key Takeaways (3 min)

2× Table • All answers are EVEN • End in 0, 2, 4, 6, or 8 • Go up by 2 each time • Multiplying by 2 = doubling	5× Table • All answers end in 0 or 5 • Go up by 5 each time • Odd × end in 5; even × end in 0 • Links to clock minute hand	Commutativity • $2 \times 5 = 5 \times 2 = 10$ • Order doesn't change the answer • Know one fact → know two!
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10. Home Challenge

■ At home tonight:

- Count all the $\pounds 2$ coins in your piggy bank and use the 2× table to find the total value.
- Count the fingers on every family member — use the 5× table!
- Write out the 2× and 5× tables from memory and decorate them.
- Teach one fact from each table to a younger sibling or parent.

11. Assessment Checklist

Use the following criteria to assess each student at the end of the session:

Criterion	Not yet	Developing	Achieved
Recalls 2× facts (1–10) fluently	☐	☐	☐
Recalls 5× facts (1–10) fluently	☐	☐	☐
Identifies even-number pattern in 2× table	☐	☐	☐
Identifies 0/5 ending pattern in 5× table	☐	☐	☐
Applies multiplication to word problems	☐	☐	☐
Understands commutativity	☐	☐	☐
Self-corrects using patterns (error-spotting)	☐	☐	☐

12. Preview — Next Session

★ **Session 2:** Times Tables 3 & 4 — More Magic Patterns!

Topics: Building the 3× table via repeated addition; the 4× table as double-double; real-life contexts; quick-fire mixed drill of 2×, 3×, 4× and 5×.

Shabash! ■ You're a Times Tables Star!
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